



ज्ञानेन विकासः

IIIT DHARWAD

PROJECT PROPOSAL

Digital Banking for Inclusion and Sustainability

A web-based platform for financial inclusion and carbon-conscious banking

A Mini Project Proposal

Platform: Responsive Website

Stack: React / Next.js • Node.js or Django REST • PostgreSQL / MongoDB • scikit-learn

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1. Project Idea (In Brief)

This project is a web-based digital banking platform that solves two problems at once. First, it brings basic banking services — accounts, savings, and small loans — to people who are currently unbanked or underbanked, such as rural users, daily-wage workers, and people with low digital literacy or no credit history. Second, it makes banking environmentally conscious, by tracking the estimated carbon footprint of a user's spending and rewarding eco-friendly financial habits.

In short: “banking that reaches everyone, and banking that's good for the planet.” The platform is delivered as a responsive website rather than a native mobile app, which keeps the project accessible on both desktop and low-end mobile browsers, removes app-store installation friction for demo purposes, and lets the team focus engineering effort on one shared codebase instead of maintaining separate iOS/Android builds.

The application will be a working prototype (not connected to real bank networks) that simulates transactions using a mock database, so the focus stays on demonstrating the concept, the user experience, and the underlying logic — which is the expected and accepted scope for a student mini project.

2. Problem Statement

- Financial exclusion: A large share of the population — rural residents, low-income workers, people without proper ID or a smartphone — remain outside the formal banking system because of complicated onboarding, lack of nearby branches, language barriers, and most importantly, no credit history to qualify for loans.
- Environmental blindness of banking: Most banking platforms today give users zero visibility into the environmental impact of how they spend money, and banking itself still relies heavily on paper statements, physical forms, and branch visits.

Our project addresses both gaps within a single, unified platform.

3. Objectives

- Design a simple, accessible web banking interface that lower-literacy and first-time users can operate confidently on any device, including older or low-spec ones.
- Build an alternative credit-scoring mechanism so users with no prior credit history can still qualify for small loans.
- Implement a carbon footprint tracker and a green-rewards system to encourage sustainable spending habits.
- Demonstrate the full user journey — onboarding, transacting, saving, borrowing — through a working simulated prototype.
- Add a second data-driven layer (fraud/anomaly detection) so the platform showcases more than one applied ML use case.

4. Market Context and Why This Problem Still Matters

Before finalising the feature list, we looked at how large the two problems actually are today, so the project targets real gaps rather than assumed ones.

4.1 The scale of financial exclusion

- The World Bank's Global Findex Database 2025 (“Connectivity and Financial Inclusion in the Digital Economy,” based on a 2024 survey of ~145,000 adults across 141 economies) found that 1.3 billion adults worldwide are still unbanked, and over half of them are concentrated in just eight countries, including India.

- Within the unbanked population, 55% are women, 52% belong to the poorest 40% of households, and 62% have only primary education — confirming that exclusion is concentrated among exactly the low-literacy, low-income users this proposal targets.
- India itself has made rapid progress on account ownership (up to 89% in 2024 from just 35% in 2011, largely credited to Jan Dhan Yojana and biometric-linked onboarding), but a meaningful share of these accounts remain inactive or underused — meaning the harder problem is no longer just opening an account, but making it usable enough for someone with low literacy or no credit history to actually save, borrow, and transact.
- Notably, of the adults worldwide who remain unbanked, roughly 900 million already own a mobile phone and about 530 million own a smartphone — which is the key justification in this proposal for building a responsive web platform rather than assuming users only have basic feature phones.

4.2 Why category-based carbon tracking is a credible approach

On the sustainability side, spend-based carbon accounting (estimating emissions from what money was spent on, rather than measuring the physical activity directly) is not a shortcut invented for this project — it is a recognised methodology (GHG Protocol Scope 3.1) used by real emission-factor providers such as Climatiq, and by consumer fintechs such as Doconomy and Aspiration, both cited in Section 8. This gives us confidence that the “worked example” approach used in Section 6 is methodologically defensible, even at prototype scale.

5. Core Banking Module (Foundation)

This is the base layer every other feature sits on top of.

- **User Registration & Login:** Simple sign-up with phone number/basic ID, secure authentication (OTP-based, session/JWT tokens).
- **Account Dashboard:** Shows balance, recent transactions, and quick-action buttons — built as a responsive web dashboard that reflows cleanly on mobile browsers.
- **Simulated Money Transfer:** Send/receive money between users using a mock wallet system (balances updated in our own database — no real bank network involved, since that requires licenses beyond student scope).
- **Transaction History:** A clear, filterable log of all activity, searchable by date, category, or amount.
- **Bill Payment / Recharge (simulated):** Mock utility bill and recharge payments.

6. Financial Inclusion Module

This is what makes the platform usable and useful for unbanked/underbanked users.

- **Simplified Onboarding:** Minimal form fields, icon-based navigation, multilingual interface, and voice-guided sign-up (via the browser's built-in speech APIs) so literacy is not a barrier.
- **Alternative Credit Scoring:** Since underbanked users have no traditional credit history, we calculate a “trust score” (0–100) using their in-app behavior instead — savings regularity, on-time simulated bill payments, and consistent transaction activity.
- **Micro-Loans:** Loan eligibility and amount are determined by the trust score, giving users who would be rejected by traditional banks a path to small, manageable credit.
- **Micro-Savings Goals:** Small, goal-based savings (e.g., saving for school fees) with visual progress tracking.
- **Financial Literacy Section:** Short lessons and quizzes (in local language) on budgeting, saving, and avoiding fraud, with badges for completion.

- **Voice Assistant:** A mic-based assistant (Web Speech API) so users can say things like “check my balance” instead of typing/reading.

Additional features (added for depth)

- **Family / Group Accounts:** Let multiple family members share visibility into one account — useful for households where one person manages money for everyone.
- **Nearby Agent / Kiosk Locator:** A simulated directory/map of local banking correspondents or agents for cash deposit or withdrawal, since many unbanked users still need a human touchpoint even when using a website.
- **Community Trust Circles:** Small peer groups where members can vouch for each other, feeding into the trust score — mirrors how real microfinance self-help groups work.
- **Financial Literacy Chatbot:** A simple rule-based or LLM-powered chatbot answering basic banking and finance questions in the local language, complementing the literacy lessons.

7. Sustainability Module

This is what makes the platform environmentally conscious.

- **Carbon Footprint Tracker:** Every transaction is tagged with a category (fuel, travel, groceries, electronics, etc.). Each category has a pre-defined emission factor, and the app calculates and displays an estimated CO2 footprint for the user's spending, with a monthly trend graph.
- **Green Rewards System:** Users earn “green points” for low-carbon spending choices (public transport, e-bill payments, local produce), visible on a small leaderboard/rewards page.
- **Paperless Metrics:** The app shows users a fun, cumulative stat like “you've saved X sheets of paper by going digital,” reinforcing the value of paperless banking.
- **Sustainability Score:** A simple score combining the share of green-category spending and green points earned, shown alongside the trust score.

Additional features (added for depth)

- **Carbon Offset Marketplace:** Users can spend green points to fund small offset projects (tree planting, clean cookstoves), shown as cumulative impact stats.
- **Peer Comparison Leaderboard:** Shows users how their footprint compares with others in their village/town or peer group, encouraging friendly competition.

What “CO2 footprint” actually means here

The carbon footprint in this app is an estimated figure based on spending category, not a lab-measured value. It works like this:

- Every transaction is tagged with a category — fuel, travel, groceries, electronics, dining, and so on.
- Each category has a pre-set emission factor — a fixed number that says, for example, “spending ₹100 on fuel roughly equals X kg of CO2.”
- The app multiplies spending in that category by the emission factor to get an estimated CO2 value for that transaction.
- These values are summed across the month into a total footprint, shown as a trend graph.

This is a proxy estimate, not a scientifically precise measurement — real emissions depend on exact product, distance travelled, and energy source, which is beyond the scope of a mini project. Real apps like Doconomy and Aspiration use the same category-based approach. Emission factor numbers for this project

can be sourced from publicly available data such as India's Central Electricity Authority (CEA) figures for electricity, or general published lifecycle-emission estimates for fuel and travel.

8. Standout (“Wow”) Features

- **1. ML-based Trust Score:** We will implement the alternative credit-scoring system using a machine learning model (trained on simulated/sample user behavior data) rather than a simple fixed formula. This turns the inclusion feature into a data-driven system and is the strongest technical highlight of the project.
- **2. Fraud / Anomaly Detection:** A lightweight anomaly-detection model flags unusual transaction patterns (sudden large transfers, atypical timing, rapid repeated transactions). This adds a second applied ML use case beyond credit scoring, and doubles as a financial-literacy safety feature.

9. Similar Real-World Systems

These platforms validate that the core ideas in this proposal are already proven in the market, not purely hypothetical:

Worked example: ₹2,000 spent on fuel this month, with an emission factor of 2.5 kg CO₂ per ₹100 spent, gives an estimated footprint of $2000 \times 0.025 = 50$ kg CO₂ from fuel alone. Repeat for every category and sum the results for the monthly total.

Alternative credit scoring

- **Tala** — operates in Kenya, the Philippines, Mexico, and India; scores creditworthiness using phone usage, transaction patterns, and behavioral data instead of a credit bureau history, very close to the “trust score” concept in this proposal.
- **Branch International** — uses smartphone data (contacts, SMS, app usage) to score users instantly and disburse microloans with no paperwork.
- **India Stack / Jan Dhan Yojana** — the real government-backed push for financial inclusion in India, with simplified biometric-linked onboarding that enabled millions of previously unbanked citizens to open accounts.
- **Kiva** — a global micro-lending platform connecting lenders to underserved borrowers, relevant as a reference for the trust-based micro-loan angle.

Carbon tracking in banking

- **Doconomy** — a Swedish fintech that partnered with Mastercard on a card tracking the CO₂ footprint of every purchase, closely matching this proposal's carbon tracker.
- **Aspiration** — a US neobank that scores the environmental impact of purchases (“Aspiration Impact Measurement”) and plants trees for qualifying purchases, a direct parallel to the green rewards system here.
- **Commonwealth Bank of Australia** — added a carbon footprint tool directly into its mainstream banking app, showing that incumbent banks, not just startups, are adopting this feature.

Low-literacy / voice-first banking

- **Eko India** — enabled banking through basic phones and local kirana-store agents for underbanked users, a real example of banking that doesn't require a smartphone or high literacy.
- **PayJoy** — provides device financing with phone-based repayment tracking, in the same spirit as this proposal's behavior-based trust score.

10. Additional Proposed Enhancements (Research-Backed)

To push the project beyond a purely conceptual prototype, we propose two further additions. Both were chosen because they map to genuine, current regulatory and industry practice rather than being invented for the proposal, and both are realistic to implement within the mini-project timeline.

10.1 Explainable Trust Score (“Why is my score X?”)

- The RBI's Digital Lending Directions, 2025 (which updated the 2022 guidelines) push lenders toward disclosing the basis of automated credit decisions to borrowers, and industry commentary points to explainability and disclosure of credit-scoring methodology as a growing regulatory expectation for AI-driven lending in India.
- Implementation: alongside the numeric trust score, show a simple factor-contribution breakdown — for example, “+15 for regular savings, +10 for on-time bill payments, –10 for irregular activity” — derived from the feature weights/coefficients that the scikit-learn model already computes internally.
- This is technically lightweight (no new model is needed, only a readable view of existing feature importances), but it visibly demonstrates awareness of explainable-AI and fair-lending practice, which is a strong talking point for evaluation.

10.2 Standards-Based Carbon Calculation via a Spend-Based Emission API

- Instead of relying only on a small internal table of self-defined emission factors, the carbon tracker can optionally call a spend-based emission-factor API such as ClimaTiq, which maps transaction categories to emission factors validated against the GHG Protocol and returns an auditable CO2e estimate per transaction.
- Practical approach for a mini project: use a free developer-tier API key for a handful of common categories (fuel, electricity, travel, groceries), and fall back to the existing static emission-factor table (Section 7) for categories outside the free quota or when working offline for the demo.
- This keeps the worked-example methodology already described in this proposal, but backs it with a credible, standards-aligned data source instead of only self-defined numbers — something that is easy to point to and justify at the review or viva.

11. Machine Learning Methodology

This section lays out, in more depth, how the two “wow” features from Section 8 would actually be built, based on approaches reported in recent credit-scoring and fraud-detection literature.

11.1 Trust score model

- Baseline model: a logistic regression “scorecard” — the same structure traditional bank scorecards use — trained on simulated behavioral features (savings regularity, on-time bill-payment ratio, transaction consistency, account age, group/family activity). Logistic regression is chosen as the baseline because its coefficients are directly interpretable (each feature has one fixed, signed weight), which keeps the explainability feature in Section 10.1 simple and auditable.
- Stronger model (stretch goal): a gradient-boosted tree model (e.g., XGBoost or scikit-learn's HistGradientBoostingClassifier) trained on the same features. Published comparisons consistently show gradient-boosted trees outperforming logistic regression on credit-scoring tasks by capturing non-linear interactions between features (for example, “regular savings only helps the score if bill payments are also on time”).
- Because tree-based models are harder to explain directly, if we use one we will generate the factor-contribution breakdown in Section 10.1 using each feature's SHAP (SHapley Additive exPlanations) value instead of a raw coefficient — SHAP is the most commonly used post-hoc explanation method in the credit-scoring literature specifically because it works on top of black-box models like XGBoost without changing how they are trained.

- We will keep the feature set intentionally free of protected attributes (gender, religion, caste, etc.) so the trust score cannot proxy-discriminate — this mirrors how published fairness studies test credit models by removing sensitive attributes and checking that accuracy does not depend on them.

11.2 Fraud / anomaly-detection model

- Because a student prototype will not have real labelled fraud cases, we treat this as an unsupervised anomaly-detection problem rather than a supervised classification problem — the same choice made in the fraud-detection literature when genuine fraud examples are scarce.
- Primary technique: Isolation Forest (available directly in scikit-learn) — it isolates points by randomly partitioning the feature space, so transactions that are easy to isolate (very large amount, unusual hour, rapid repeated transfers) get a high anomaly score. It is fast, needs no labelled data, and is widely used as a first-pass fraud filter in published comparisons.
- Optional stretch comparison: a simple autoencoder (a small neural network trained only on normal transactions) that flags a transaction as anomalous when it cannot reconstruct it well (high reconstruction error). Literature comparing the two shows autoencoders can catch subtler anomalies than Isolation Forest but need more data and tuning — reasonable to mention as a “future work” comparison rather than commit to building both within 10 weeks.
- Flagged transactions are not auto-blocked in the prototype; they are surfaced to the user (“was this you?”) and logged, which keeps the feature demonstrable without needing a real fraud-response workflow.

12. System Architecture (High-Level)

The platform follows a standard three-tier web architecture, kept deliberately simple so the team can build and demo it within the timeline in Section 16.

12.1 Layers

- Client layer: React/Next.js single-page application, responsible for the dashboard, onboarding flow, voice assistant (via the Web Speech API), and multilingual UI switching.
- API layer: a REST API (Node.js/Express or Django REST Framework) exposing endpoints for authentication, accounts, transactions, trust score, carbon score, and admin/analytics — with the ML components (trust score, anomaly detection) exposed as internal service calls the API layer invokes after each simulated transaction.
- Data layer: PostgreSQL for structured, relational data (users, accounts, transactions, loans) and, optionally, MongoDB for less structured data such as chatbot conversation logs or literacy-quiz content.
- ML layer: a small Python service (scikit-learn models loaded from a serialized file) that the API layer calls synchronously for trust-score and anomaly-score requests; kept as a separate module so the models can be retrained/replaced without touching the main API code.

12.2 Key data flow

- A simulated transaction is created → written to PostgreSQL → tagged with a spending category → the carbon module computes a CO2e estimate (Section 7, backed by Section 10.2) → the anomaly-detection module scores the transaction → the trust-score module recalculates the user's score if the transaction affects a scoring feature → the dashboard reflects all updated numbers on next load.

12.3 Security and data handling (prototype scope)

- OTP-based login with session/JWT tokens (Section 5); passwords/OTPs hashed, never stored in plaintext.
- Because this is a simulated prototype (Section 1), no real financial credentials, card numbers, or government ID numbers are collected or stored — onboarding only uses a mock ID field, which keeps the project outside the compliance scope of real banking regulation while still demonstrating the UX of ID-based onboarding.

13. Risks, Limitations, and Mitigation

- **Simulated data may not reflect real user behavior:** Synthetic/sample data used to train the trust-score and anomaly models may not capture real-world patterns. Mitigation: generate simulated data using distributions grounded in the cited real-world sources (e.g., typical savings/repayment patterns reported for Tala/Branch-style scoring), and clearly label the prototype's numbers as illustrative, matching how the carbon-footprint methodology in Section 7 is already framed.
- **Emission factors are estimates, not measurements:** As already noted in Section 7, spend-based carbon figures are proxies. Mitigation: source factors from credible public datasets (CEA, published lifecycle estimates) or the ClimaTiq API (Section 10.2), and present the number as “estimated CO₂e” everywhere in the UI, never as an exact figure.
- **Class imbalance in fraud detection:** Genuine fraud is rare, which is exactly why supervised classification is hard without real labelled data (Section 11.2). Mitigation: use unsupervised anomaly detection (Isolation Forest) rather than a classifier that needs balanced fraud/non-fraud examples.
- **Explainability adds UI complexity:** Showing a factor-contribution breakdown (Section 10.1) requires additional design work beyond a plain score. Mitigation: scope it to a simple bar-style breakdown of the top 3–4 contributing factors rather than a full SHAP visualization, which keeps it buildable within the timeline.
- **Team scope vs. timeline:** With two ML components, a voice assistant, and a multilingual UI, feature creep is a real risk for a 10-week mini project. Mitigation: the timeline in Section 16 sequences the core banking module first, treats the ML enhancements in Section 10 as an explicit Week 9 add-on, and treats the second anomaly-detection technique in Section 11.2 (autoencoder) as optional future work rather than a committed deliverable.

14. Evaluation Plan

Since the platform is a simulated prototype rather than a production system, evaluation focuses on whether each module works as intended and whether the ML components behave sensibly on test data, rather than on live business metrics.

- Core banking module: manual and scripted test cases for signup, login, simulated transfer, and transaction history (correct balances after N simulated transactions, correct filtering by date/category/amount).
- Trust-score model: standard classification metrics on a held-out slice of the simulated dataset (accuracy, precision/recall, and AUC-ROC, which is the metric most commonly reported in the credit-scoring literature cited in Section 11.1), plus a manual sanity check that the top contributing factors shown to a user (Section 10.1) match intuition (e.g., a user with irregular savings should not receive a high positive contribution for savings regularity).
- Anomaly detection: evaluate the Isolation Forest's contamination-rate assumption against the simulated dataset (what fraction of transactions it flags) and spot-check flagged transactions against the synthetic “known anomalies” injected into the test data.
- Carbon tracker: cross-check a handful of worked examples (as in Section 9) by hand against both the internal static table and the ClimaTiq API output, to confirm the two sources stay within a reasonable range of each other.
- Usability (inclusion angle): a short informal usability pass with 3–5 volunteers unfamiliar with the platform, checking whether the simplified onboarding and voice assistant (Section 6) can be completed without help — directly testing the low-literacy objective from Section 3.

15. Proposed Tech Stack (Web-First)

Layer	Technology
Frontend	React / Next.js — responsive website, works on desktop and low-end mobile browsers
Backend	Node.js (Express) or Django REST Framework
Database	PostgreSQL or MongoDB
ML Component	Python (scikit-learn) for credit-scoring and anomaly-detection models
Voice Assistant	Web Speech API (browser-native, no app SDK needed)
Extras	Twilio (simulated SMS banking) — optional stretch goal

16. Tentative Timeline

Phase	Deliverable
Week 1–2	Requirement study, wireframes, database design
Week 3–5	Core banking module: auth, accounts, simulated transfers
Week 6–7	Inclusion module: onboarding, trust score, micro-loans
Week 8	Sustainability module: carbon tracker, green rewards
Week 9	ML credit-scoring and fraud-detection model integration, plus explainability view and carbon-API integration (Section 10)
Week 10	Testing against the evaluation plan (Section 14), polishing, final report and demo

17. Expected Outcome

A working web-based prototype that demonstrates the complete user journey of an unbanked user — from simplified onboarding, through building a trust score via everyday app behavior, to receiving a micro-loan — while also giving every user visibility into, and rewards for, the environmental impact of their spending.

The project will show that digital banking can advance financial inclusion and environmental sustainability at the same time, through one connected, browser-accessible platform — backed by real-world precedents in both the fintech-for-inclusion space (Tala, Branch, India Stack) and the carbon-conscious banking space (Doconomy, Aspiration, CommBank), strengthened by the explainability and standards-aligned carbon calculation additions in Section 10, and validated against the evaluation plan in Section 14.

18. References

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- KNIME / Medium (2022). Fraud Detection using Random Forest, Neural Autoencoder, and Isolation Forest techniques.
- Tala, Branch International, Kiva, Eko India, PayJoy — public product documentation, referenced in Section 9 for alternative credit scoring and low-literacy banking precedents.

-----Thank You-----